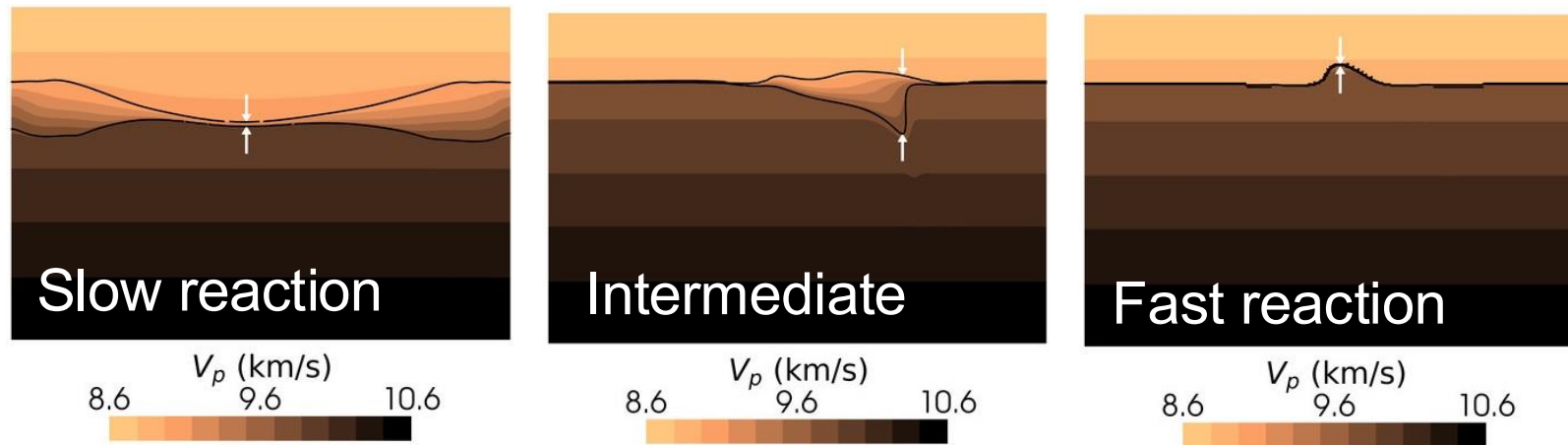


Beyond Equilibrium

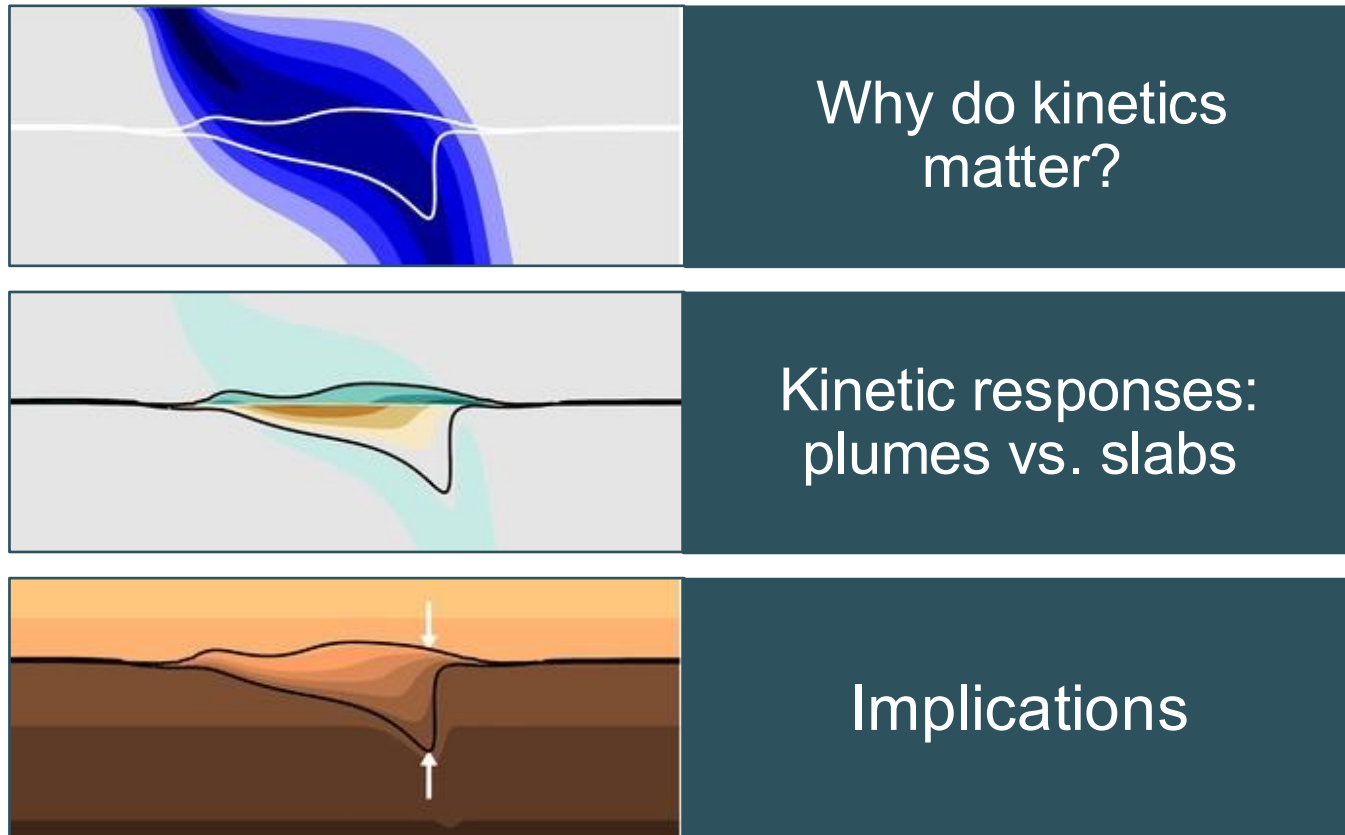
Kinetic Thresholds and Rheological Feedbacks Create a Potentially Complex 410 in Slab Regions



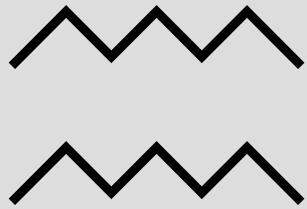
B. Kerswell
J. Wheeler
R. Gassmöller
J.H. Davies
I. Papanagnou
S. Cottaar



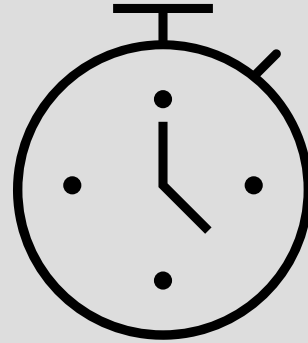
We simulated plumes and slabs to explore how kinetics and slab strength shape the 410 km discontinuity



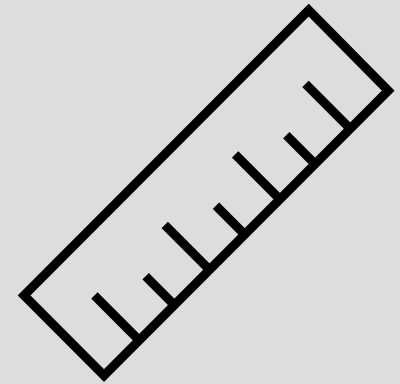
The 410 varies globally, yet reaction kinetics remain largely unexplored in dynamic mantle flow simulations



The 410 appears sharp in some settings and broad or absent in others

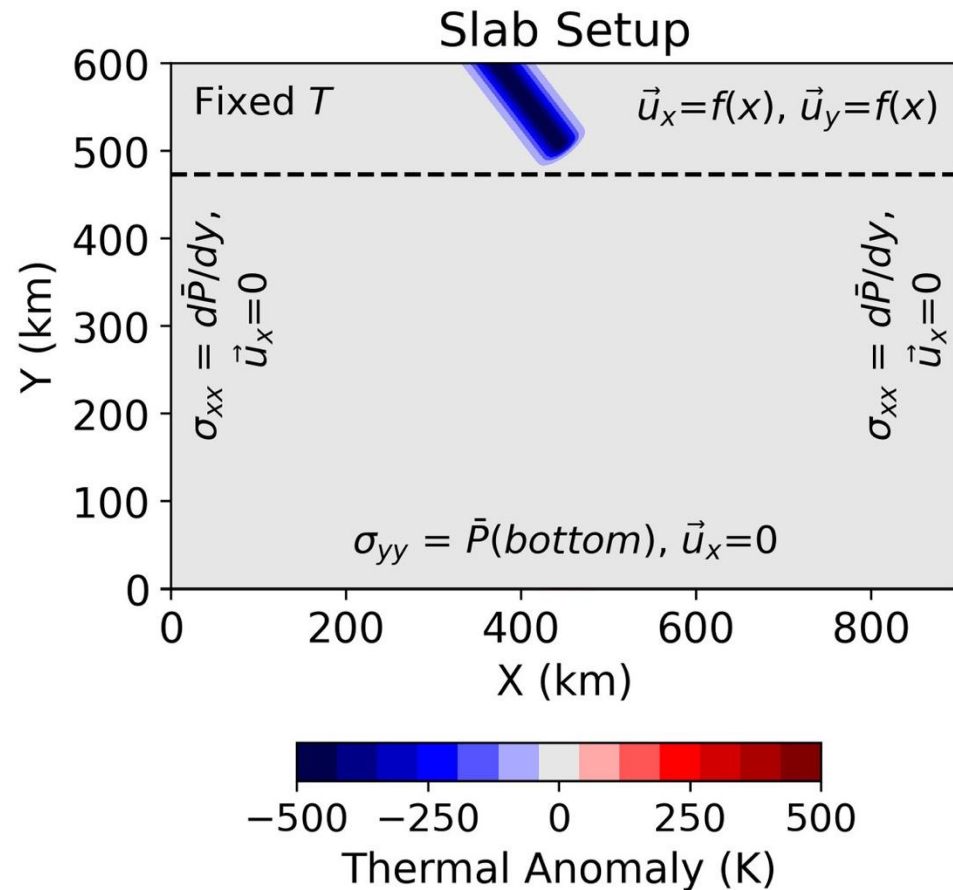


Most mantle flow models assume instant mineral transformation



Published kinetic parameters span 7 orders of magnitude

We coupled a grain-scale olivine-wadsleyite growth model to compressible mantle flow simulations, varying kinetic conditions across 7 orders of magnitude



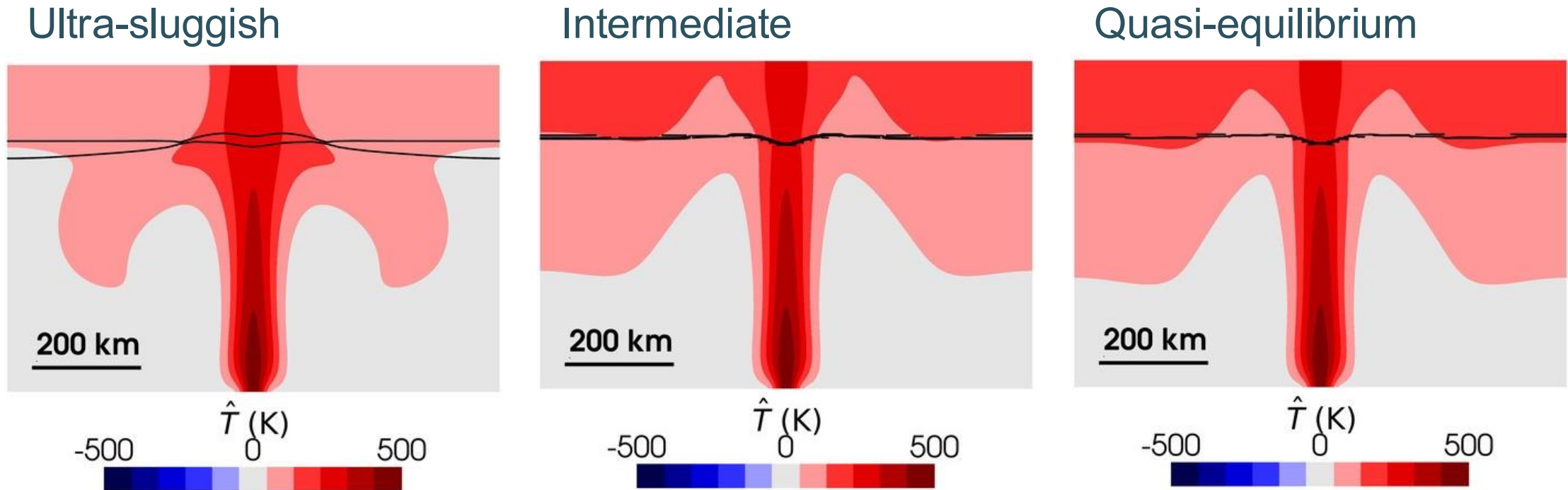
Wadsleyite growth rate model:

$$\dot{X} = Z T \exp\left(-\frac{H + PV}{RT}\right) \left[1 - \exp\left(-\frac{\Delta G}{RT}\right)\right] (1 - X)$$

Rheological model:

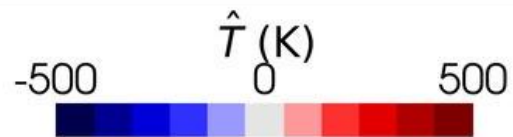
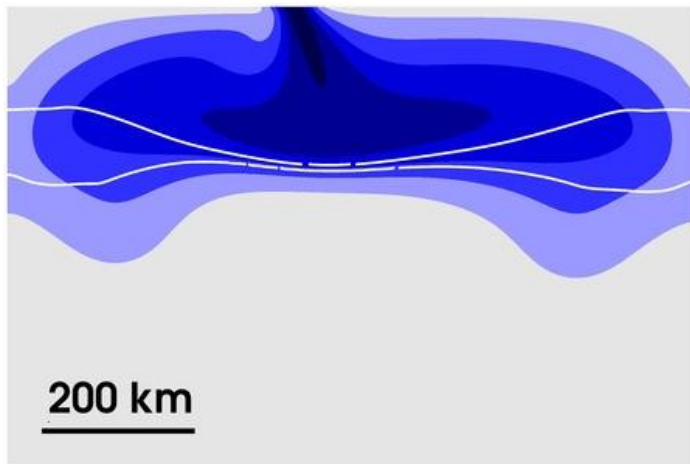
$$\eta = \bar{\eta} \exp\left(-B \frac{\hat{T}}{\bar{T}}\right)$$

In plumes, the 410 stays sharp (2–3 km wide) across all tested reaction rates: temperature dominates

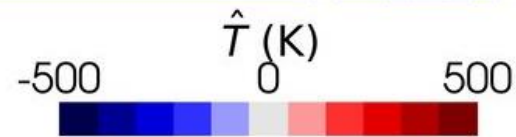
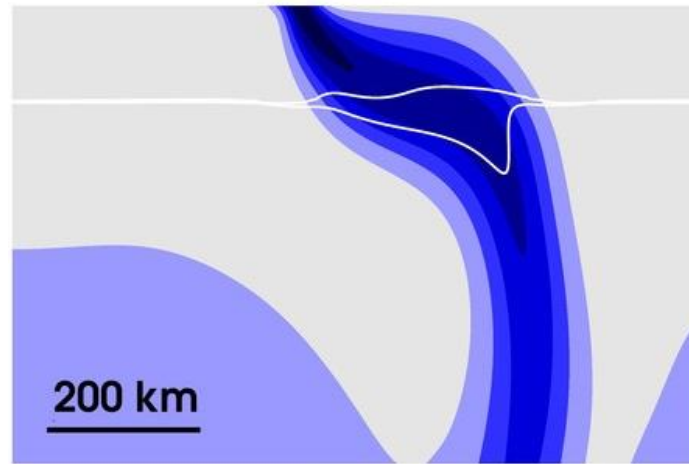


**In slabs, three kinetic regimes determine 410 structure:
stagnant/displaced by ~100 km , broad/deep, or sharp/uplifted**

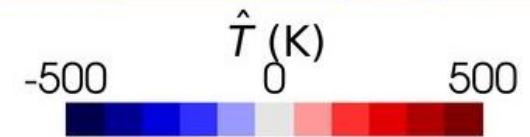
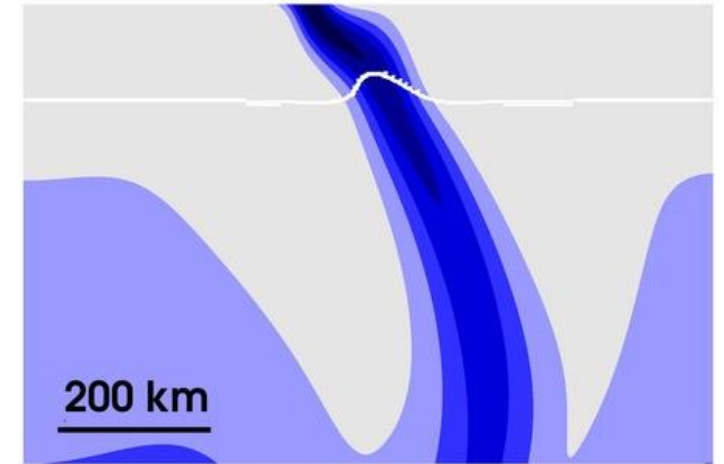
Ultra-sluggish



Intermediate

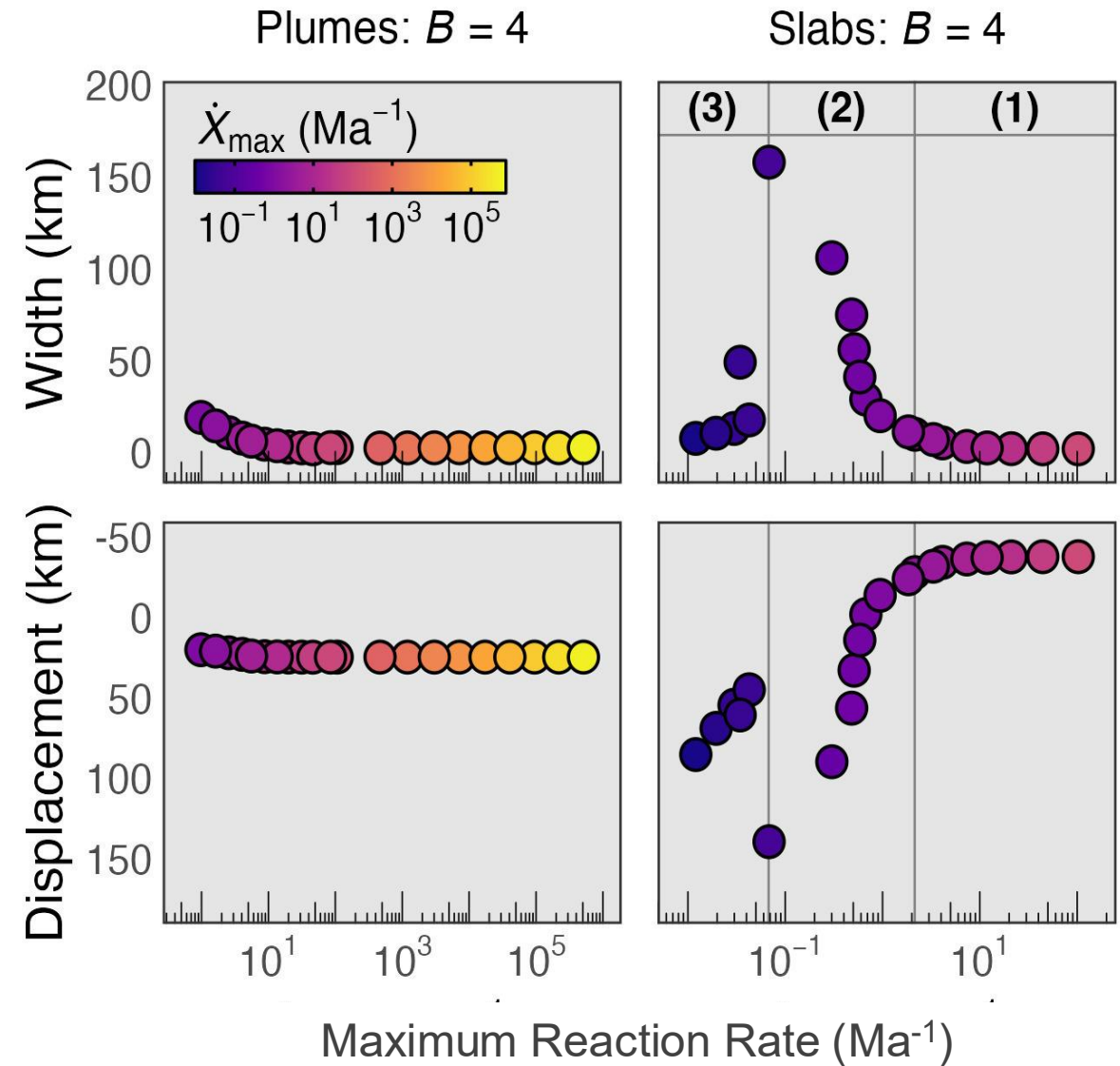


Quasi-equilibrium



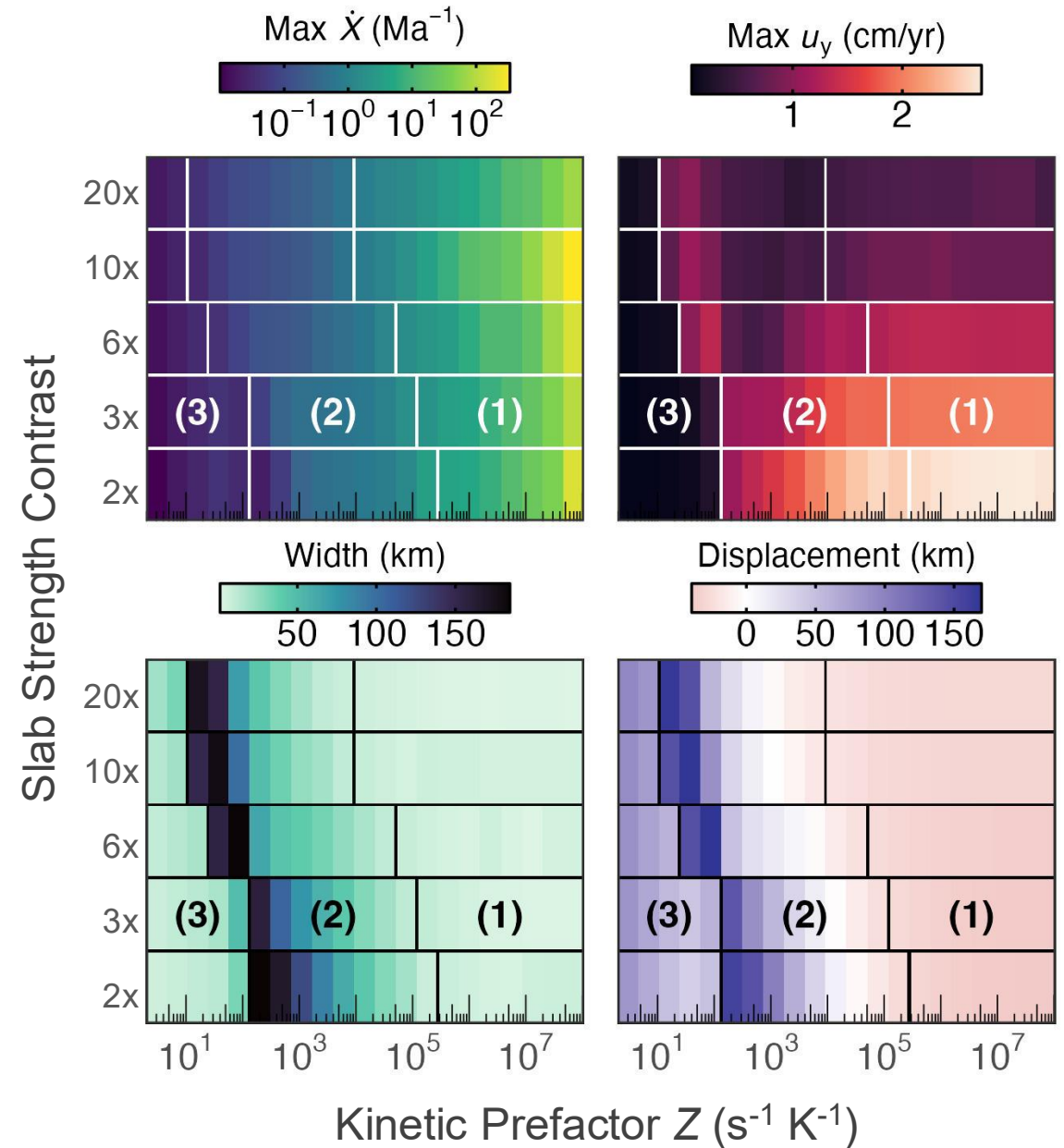
Plume 410 structure is kinetically insensitive

Slab 410 structure changes non-linearly through three kinetic regime



Slab strength shifts regime boundaries

Under the same kinetic conditions: weak slabs amplify metastability; stronger slabs favor equilibrium



In summary: reaction rates and slab strength jointly control the 410 km discontinuity

Plumes produce sharp 410s; slabs exhibit three kinetic regimes controlled by reaction rate

Slab strength modulates kinetic effects: strong slabs resist stagnation; weak slabs descend fast, amplifying disequilibrium and broadening the 410

The 410 in subduction zones *can* constrain effective reaction rates, but independent constraints are required

